Radiative decay rates for localized plasmon modes in coated systems

This is an extension of the set of notes found here:
https://hackmd.io/@aligho/Bkq90Xxybx (https://hackmd.io/@aligho/Bkq90Xxybx)

Appendix 1: Method 1 for spherical systems

The potential and field within the sphere are given by (for the dipolar mode):

$$\Phi_<(r,\theta) = br\cos(\theta) \to \mathbf{E}_< = -b\Big[\cos(\theta)e_r - \sin(\theta)e_\theta\Big] = -be_z$$

and the potential and field outside the sphere are given by

$$\Phi_>(r,\theta) = ba^3 r^{-2}\cos(\theta) \to \mathbf{E}_>(r,\theta) = b(a/r)^3\Big[2\cos(\theta)e_r + \sin(\theta)e_\theta\Big].$$

The induced charge density, $\rho(\theta)$ (charge per unit area) on the coating is given by (using notation from Localized plasmons in graphene-coated nanospheres (https://link.aps.org/pdf/10.1103/PhysRevB.91.125414)):

$$\rho(\theta) = (-i/\omega)\nabla\cdot(\sigma(\omega)b\sin(\theta)e_\theta) = (-2i/\omega)\cos(\theta)\sigma(\omega)b/a \equiv -2g(\omega)b\varepsilon_0\cos(\theta),$$

which means that, from the change in the radial component of the displacement field over the boundary at $r=a$, we must have $2+\varepsilon_m = -2g(\omega)$ (which is just the quasistatic surface plasmon dispersion–see https://hackmd.io/@aligho/ByCdz6A3ex (https://hackmd.io/@aligho/ByCdz6A3ex)). We henceforth denote the frequency at which this condition is met as $\omega=\omega_0$.

In order to use **Method 1**. We must now use the above field and charge density profile to calculate the radiated power and the stored electromagnetic energy. In order to do the former, we calculate the **dipole moment** and use the standard formula for time-averaged radiated power of an electric dipole. To do the latter, we use the Brillouin formula.

The dipole moment

The total dipole moment is given by:

$$-b\varepsilon_0(\varepsilon_m-1)\frac{4}{3}\pi a^3 - 2\pi a^3\int_0^\pi 2bg(\omega_0)\varepsilon_0\cos^2(\theta)\sin(\theta)\mathrm{d}\theta = \\ -b\varepsilon_0 a^3\left[\frac{4}{3}\pi(\varepsilon_m-1)+(8\pi/3)g(\omega_0)\right] = 4b\varepsilon_0\pi a^3,$$

where the first term corresponds to the bulk polarization density and the second term corresponds to the dipole moment of the surface charge. From the dipole moment, we obtain the total radiated power for the plasmon at frequency ω_0:

$$P(\omega_0) = \frac{\mu_0\omega_0^4(16\pi^2 b^2\varepsilon_0^2)a^6}{12\pi c} = \frac{4\pi a^3}{3}(a\omega_0/c)^3\varepsilon_0 b^2\omega_0$$

We now calculate the stored electric energy $U(\omega_0) = U_{\mathrm{bulk}} + U_{\mathrm{surface}} + U_{\mathrm{environment}}$, which is given by the following expression:

$$U(\omega_0) = \frac{1}{4}\varepsilon_0\varepsilon_m\left[\frac{4}{3}\pi a^3\right]|\mathbf{E}_<|^2 + \frac{1}{4}\varepsilon_0(2\pi)\int_a^\infty\int_0^\pi |\mathbf{E}_>(r,\theta)|^2 r^2\sin(\theta)\mathrm{d}\theta\mathrm{d}r + \\ \frac{1}{4}\varepsilon_0\left[1-i\sigma(\omega_0)/\omega_0 d\varepsilon_0\right](2\pi)\int_{a-d/2}^{a+d/2}\int_0^\pi |\mathbf{E}_<\cdot e_\theta|^2\mathrm{d}\theta r^2\mathrm{d}r,$$

where the first term corresponds to U_{bulk}, the second term corresponds to $U_{\mathrm{environment}}$ and the last term corresponds to U_{surface}. In the expression for U_{bulk}, we used the fact that the electric field in the sphere is (quasistatically) uniformly polarized in the z direction. In the expression for U_{surface}, we used the Brillouin formula with the effective surface permittivity:

$$\varepsilon_s(\omega) = 1 + \frac{i\sigma(\omega)}{\omega d\varepsilon_0} \rightarrow \frac{\partial(\omega\varepsilon_s(\omega))}{\partial\omega} = 1 - \frac{i\sigma(\omega)}{\omega d\varepsilon_0},$$

where d is the effective thickness of the surface layer (which drops out in the end). We calculate the surface contribution to the energy:

$$U_{\text{surface}} = \frac{1}{4}\varepsilon_0\Big[1 - i\sigma(\omega)/\omega d\varepsilon_0\Big](2\pi)\int_{a-d/2}^{a+d/2}\int_0^{\pi}|\mathbf{E}_< \cdot e_\theta|^2 \sin(\theta)\mathrm{d}\theta r^2 \mathrm{d}r \approx$$
$$-\frac{1}{4}\varepsilon_0\frac{i\sigma(\omega)}{\omega d\varepsilon_0}\frac{2\pi}{3}\Big[(a+d/2)^3 - (a-d/2)^3\Big]\frac{4}{3}|\mathbf{E}_<|^2 \approx -\frac{2\pi}{3}\varepsilon_0 g(\omega)a^3|\mathbf{E}_<|^2$$

We add this to the contribution from the energy stored in the sphere to give:

$$U_{\text{surface}} + U_{\text{bulk}} = \frac{1}{3}\pi a^3 b^2\varepsilon_0\bigg[-2g(\omega) + \varepsilon_m\bigg] = \frac{2}{3}\pi a^3 b^2 \varepsilon_0(\varepsilon_m + 1)$$

Now we need only calculate the energy stored outside the sphere:

$$U_{\text{environment}} = \frac{1}{4}\varepsilon_0(2\pi)\int_a^{\infty}\int_0^{\pi}|\mathbf{E}_>(r,\theta)|^2 r^2\sin(\theta)\mathrm{d}\theta\mathrm{d}r =$$
$$\frac{\varepsilon_0\pi}{2}b^2a^6\bigg[\frac{1}{3a^3}\bigg]\int_0^{\pi}\Big[4\cos^2(\theta) + \sin^2(\theta)\Big]\sin(\theta)\mathrm{d}\theta = \frac{\varepsilon_0\pi b^2 a^3}{6}\bigg[\frac{8}{3} + \frac{4}{3}\bigg] =$$
$$\frac{2}{3}\pi a^3 b^2\varepsilon_0$$

Therefore, the total energy is given by: $U = \frac{2}{3}\pi a^3 b^2 \varepsilon_0(\varepsilon_m + 2)$ and the decay rate will be given by:

$$-2\gamma \equiv \frac{P(\omega_0)}{U(\omega_0)} = \frac{\frac{4\pi a^3}{3}(a\omega_0/c)^3\varepsilon_0 b^2\omega_0}{\frac{2}{3}\pi a^3 b^2\varepsilon_0(\varepsilon_m + 2)} = \frac{2\omega_0(a\omega_0/c)^3}{\varepsilon_m + 2}$$

Where $\omega = \omega_0 + i\gamma$ is the complex plasmon frequency.

Appendix 2: Method 2 for spherical systems

The exact solution for the decay rate is given by $\omega = \omega_0 + i\gamma$ which solves the equation (see, for instance, Localized plasmons in graphene-coated nanospheres (https://link.aps.org/pdf/10.1103/PhysRevB.91.125414)):

$$h_l^1(x_2)[x_1 j_l(x_1)]' - \varepsilon_m j_l(x_1)[x_2 h_l^1(x_2)]' - g(\omega)[x_1 j_l(x_1)]'[x_2 h_l(x_2)]' = 0,$$

Our **Method 2** equation may be written as $f(\omega) + ih(\omega) = 0$, where we define $f(\omega)$ and $h(\omega)$ as follows:

$$f(\omega) = j_l(x_2)[x_1 j_l(x_1)]' - \varepsilon_m j_l(x_1)[x_2 j_l(x_2)]' - g(\omega)[x_1 j_l(x_1)]'[x_2 j_l(x_2)]',$$
$$h(\omega) = y_l(x_2)[x_1 j_l(x_1)]' - \varepsilon_m j_l(x_1)[x_2 y_l(x_2)]' - g(\omega)[x_1 j_l(x_1)]'[x_2 y_l(x_2)]'$$

We use expansions of the spherical bessel functions (for small arguments) from https://hackmd.io/@aligho/ByCdz6A3ex (https://hackmd.io/@aligho/ByCdz6A3ex) in order to write:

$$j_1(x_1) \approx \frac{x_1}{3}, [x_1 j_l(x_1)]' \approx \frac{2x_1}{3}, [x_2 j_l(x_2)]' \approx \frac{2x_2}{3},$$
$$y_l(x_2) \approx -\frac{1}{x_2^2}, [x_2 y_l(x_2)]' \approx \frac{1}{x_2^2}$$

Which gives us the following for $f(\omega)$ and $h(\omega)$:

$$f(\omega) \approx \frac{2}{9}\Bigg[(1-\varepsilon_m) - 2g(\omega)\Bigg]x_1 x_2, h(\omega) \approx -\frac{1}{3}\Bigg[(2+\varepsilon_m) + 2g(\omega)\Bigg]\frac{x_1}{x_2^2}$$

Therefore, if we expand around the quasistatic plasmon solution (see https://hackmd.io/@aligho/ByCdz6A3ex (https://hackmd.io/@aligho/ByCdz6A3ex)), $\varepsilon_m + 2 + 2g(\omega_0) = 0$, we obtain the following condition for the plasmon frequency:

$$\frac{2}{3}\Bigg[(1-\varepsilon_m) - 2g(\omega_0) - 2ig'(\omega_0)\gamma\Bigg]\Bigg[\omega_0^3 + 3i\omega_0^2\gamma\Bigg](a/c)^3 + 2g'(\omega_0)\gamma = 0$$

Note that since $g(\omega)$ goes as ω^{-2}, we have that $g'(\omega_0) = -2g(\omega_0)/\omega_0 = (\varepsilon_m + 2)/\omega_0$, which means that, to lowest order in γ, the real part of this equation will be zero iff:

$$\gamma = -\frac{1}{3g'(\omega_0)}\left[1 - \varepsilon_m - 2g(\omega_0)\right](a\omega_0/c)^3 = -\omega_0(a\omega_0/c)^3/(\varepsilon_m + 2)$$

Therefore, if the interior is vacuum, the decay rate will be the same as an uncoated metallic sphere.

Let's also look at the imaginary part of the equation above:

$$\left[6\omega_0^2 - (4/3)\omega_0^3 g'(\omega_0)\right](a/c)^3\gamma = \left[6\omega_0^2 - (4/3)\omega_0^2(\varepsilon_m + 2)\right](a/c)^3\gamma$$

Therefore, if we insert our expression for γ, we see that we have fulfilled the boundary condition to order $(a\omega_0/c)^6$.